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Digital Logic Design

(DLD Theory)

Assignment #03.

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Assignment #03

Q#01) 4-to-1 Multiplexer

Given

$I_0 = 1$
$I_1 = 0$
$I_2 = 1$
$I_3 = 0$

Selection Line

$S_1 = 0$
$S_0 = 0$

Truth table for Selection

S_1	S_0	selection input
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Since
$S_1 S_0 = 01$
the selection input is I_1.

output = $L_1 = 0$.

Answer.

selection input = L_1

output = 0.

Question #02.

When a floor button is pressed, it must be converted into a binary code?

Answer

Encoder convert the pressed floor button into a binary code.

Decoder convert the binary code back to activate the correct floor indicator light.

Answer

Button press → Encoder

floor indicator → Decoder.

Question #03.

Security System.

First Case :-

Four cameras are connected to one monitor and one camera must be selected.

circuit used → Multiplexer (MUX).

Reason

A Multiplexer selects one input from many input and sends it to a single output.

Second Case :-

One sensor sends data to one of four alarm units.

circuit Used → Demultiplexer
(DEMUX)

Reason :- A demultiplexer sends one input to one of many outputs.

Answer

camera selection → Multiplexer

Alaram Unit selection
→ Demultiplexer.

Question #04.

Binary Subtraction

First stage :-

Subtract B From A only

circuit used → HALF subtractor.
Reason :- It subtract two bits only.

Second stage :-

Subtract B from A and considers borrow from previous stage.

Circuit Used.

FULL subtractor.

Reason :-
It subtract two bits and a borrow input.

Answer.

First stage → HALF Subtractor

Second stage → FULL subtractor

Question #05)

Digital Calculator.

(1) First Stage Circuit

HALF Adder

Reason :- Adds only two binary input A and B.

(2) Second Stage circuit.

FULL Adder

Reason :- Adds A, B, and carry

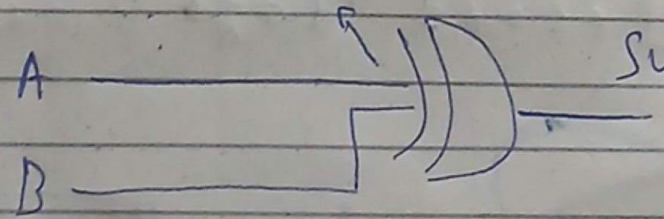
(3) Logic Diagram

HALF Adder

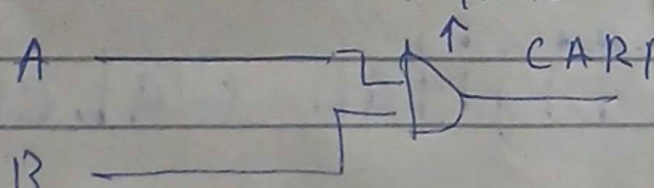
$Sum = A \oplus B$. (XOR Gate)

$Carry = A \cdot B$ (AND Gate)

Sum XOR



CARRY AND GATE



FULL Adder

Sum = A ⊕ B ⊕ Cin

carry = AB + A Cin + B Cin.

4. Truth table.

HALF ADDER.

A	B	Sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

FULL Adder.

A	B	Cin	Sum	carry.
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

5. HALF Adder output.

Given

A = 1

B = 1.

Sum = $1 \oplus 1 = 0$

carry = $1 \cdot 1 = 1$.

Answer

Sum = 0

carry = 1

FULL Adder output.

A = 1

B = 1

Carry-in = 1

Sum = $1 \oplus 1 \oplus 1 = 1$

carry = 1

Answer.

Sum = 1

Carry = 1.